

# SUPERSINGULAR CURVES AND THE TRACE OF FROBENIUS

SANTIAGO ARANGO-PÍÑEROS

Section 1 proves that an elliptic curve over  $\mathbb{F}_q$  is supersingular if and only if  $p$  divides its trace of Frobenius, and works out what this says over  $\mathbb{F}_p$ , over  $\mathbb{F}_2$  and  $\mathbb{F}_3$ , and over  $\mathbb{F}_8$ . Section 2 shows that  $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$  with  $m \mid q-1$ , and pins down  $m$  for a supersingular curve over  $\mathbb{F}_p$ . Section 3 shows that an elliptic curve over  $\mathbb{Q}$  is ordinary modulo infinitely many primes.

**Reading.** [Sil09, §§V.3–V.4], with [Sil09, §V.2] for the trace of Frobenius, [Sil09, §III.6] for the torsion subgroups and [Sil09, §III.8] for the Weil pairing. The three problems are exercises from [Sil09, Chapter V]: Problem 1.2 is Exercise 5.10, Problem 2.2 is Exercise 5.6 and Problem 3.2 is Exercise 5.11.

**Magma.** If you have no local installation, every block below runs in the free [online calculator](#).

**Standing notation.** Throughout,  $q = p^e$  is a power of a prime  $p$ . For a curve  $C$  over a field of characteristic  $p$  and  $r \geq 1$ ,  $F^r: C \rightarrow C^{(p^r)}$  is the  $p^r$ -power Frobenius morphism, which raises coordinates to the  $p^r$ -th power, and for an elliptic curve  $E$ ,  $V^r: E^{(p^r)} \rightarrow E$  is its dual isogeny, the Verschiebung. If  $E$  is defined over  $\mathbb{F}_q$ , then  $E^{(q)} = E$ , so

$$F^e: E \rightarrow E, \quad (x, y) \mapsto (x^q, y^q),$$

is the  $q$ -power Frobenius *endomorphism* of  $E$ , with dual  $V^e \in \text{End}(E)$ , and

$$a := q + 1 - \#E(\mathbb{F}_q)$$

is the [trace of Frobenius](#) of  $E$ . Here  $\text{End}(E)$  consists of the endomorphisms defined over  $\mathbb{F}_q$ . For  $m \geq 1$ ,  $E[m](\overline{\mathbb{F}}_q)$  is the group of  $m$ -torsion points of  $E$  with coordinates in  $\overline{\mathbb{F}}_q$ , and  $\deg_s \psi$  is the separable degree of an isogeny  $\psi$ .

Following [Sil09, Theorem V.3.1 and the definition after it],  $E$  is [supersingular](#) if  $E[p^r](\overline{\mathbb{F}}_q) = 0$  for one, equivalently every,  $r \geq 1$ , and [ordinary](#) otherwise. Silverman also says that a supersingular curve has “Hasse invariant 0” and an ordinary one “Hasse invariant 1”; that is the phrasing of Exercise 5.11.

**How to use this.** Allow about 35 minutes for §1, 20 for §2 and 25 for §3. In Problem 1.2 the heart is parts (1) and (5); parts (3) and (4) are hand counts to finish at home once you have looked at the tables. In Problem 2.2, part (3) is the point. Problem 3.2 is one argument in three steps; do parts (2) and (3) in the session and part (1) at home. The Magma blocks return in a few seconds.

## 1. SUPERSINGULARITY AND THE TRACE OF FROBENIUS

By Deuring’s theorem [Sil09, Theorem V.3.1], a supersingular curve is one whose Verschiebung is purely inseparable, whose endomorphism ring over  $\overline{\mathbb{F}}_q$  is an order in a quaternion algebra, and whose formal group has height 2. None of these is easy to read off from a Weierstrass equation. Over a finite field there is a criterion that is: count the rational points.

**Fact 1.1** (What we may use). Let  $E$  be an elliptic curve over  $\mathbb{F}_q$ , with  $q = p^e$ .

(1)  $F^r$  is purely inseparable of degree  $p^r$  [Sil09, Proposition II.2.11].

- (2) A purely inseparable isogeny  $\psi: E \rightarrow E'$  of degree  $p^r$  factors as  $\psi = \lambda \circ F^r$  with  $\lambda: E^{(p^r)} \rightarrow E'$  an isomorphism [Sil09, Corollary II.2.12].
- (3) A nonzero isogeny  $\psi$  satisfies  $\#\ker \psi = \deg_s \psi$ , the kernel taken on points with coordinates in  $\overline{\mathbb{F}}_q$  [Sil09, Theorem III.4.10(a)].
- (4) For  $m, n \in \mathbb{Z}$ , the endomorphism  $[m] + [n]F^e$  is separable if and only if  $p \nmid m$  [Sil09, Corollary III.5.5].
- (5)  $\hat{\psi} \circ \psi = [\deg \psi]$  and  $\psi \circ \hat{\psi} = [\deg \psi]$  for every isogeny  $\psi$  [Sil09, Theorem III.6.2(a)]; so  $V^e \circ F^e = F^e \circ V^e = [q]$ . Also  $\deg[m] = m^2$  [Sil09, Corollary III.6.4(a)].
- (6) Either  $E[p^r](\overline{\mathbb{F}}_q) = 0$  for every  $r \geq 1$ , or  $E[p^r](\overline{\mathbb{F}}_q) \cong \mathbb{Z}/p^r\mathbb{Z}$  for every  $r \geq 1$  [Sil09, Corollary III.6.4(c)].
- (7)  $(F^e)^2 - [a]F^e + [q] = 0$  in  $\text{End}(E)$  [Sil09, Theorem V.2.3.1(b)], and  $|a| \leq 2\sqrt{q}$  [Sil09, Theorem V.1.1].

**Problem 1.2** (Supersingularity and the trace of Frobenius).

This is [Sil09, Exercise 5.10]. Let  $E$  be an elliptic curve over  $\mathbb{F}_q$ .

- (1) Show that  $V^e = [a] - F^e$  in  $\text{End}(E)$ , and deduce that  $V^e$  is separable if and only if  $p \nmid a$ . Show that  $\ker V^e$  has  $\#E[q](\overline{\mathbb{F}}_q)$  elements, so that  $\deg_s V^e \in \{1, q\}$ . Conclude that

$$E \text{ is supersingular} \iff a \equiv 0 \pmod{p}.$$

- (2) Suppose  $q = p \geq 5$ . Show that  $E$  is supersingular if and only if  $\#E(\mathbb{F}_p) = p + 1$ . Where is  $p \geq 5$  used?
- (3) Every elliptic curve over  $\mathbb{F}_3$  is isomorphic to one given by  $y^2 = x^3 + a_2x^2 + a_4x + a_6$ . Magma lists the isomorphism classes, with  $\#E(\mathbb{F}_3)$ ,  $a$  and whether  $E$  is supersingular:

```
K := GF(3); reps := [];
for a2, a4, a6 in K do
  ok, E := IsEllipticCurve([K | 0, a2, 0, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;
for E in reps do
  printf "%o %o %o %o\n", aInvariants(E), #E,
    TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 0, 1, 0 ] 4 0 true
// [ 0, 0, 0, 2, 0 ] 4 0 true
// [ 0, 0, 0, 2, 1 ] 7 -3 true
// [ 0, 0, 0, 2, 2 ] 1 3 true
// [ 0, 1, 0, 0, 1 ] 6 -2 false
// [ 0, 1, 0, 0, 2 ] 3 1 false
// [ 0, 2, 0, 0, 1 ] 5 -1 false
// [ 0, 2, 0, 0, 2 ] 2 2 false
```

Which column decides supersingularity, by part (1)? Count the rational points of  $y^2 = x^3 - x + 1$  and of  $y^2 = x^3 - x - 1$  by hand, and explain why part (2) fails for  $p = 3$ .

- (4) The same over  $\mathbb{F}_2$ , where a Weierstrass equation needs all five coefficients:

```
K := GF(2); reps := [];
```

```

for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([K | a1, a2, a3, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;
for E in reps do
  printf "%o %o %o %o\n", aInvariants(E), #E,
    TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 1, 0, 0 ] 3 0 true
// [ 0, 0, 1, 1, 0 ] 5 -2 true
// [ 0, 0, 1, 1, 1 ] 1 2 true
// [ 1, 0, 0, 0, 1 ] 4 -1 false
// [ 1, 0, 1, 0, 1 ] 2 1 false

```

Count the rational points of the three supersingular curves by hand. Which coefficient decides supersingularity over  $\mathbb{F}_2$ , and what does the analogue of part (2) say for  $p = 2$ ?

- (5) Suppose  $p \mid a$ . Use (2) to write  $V^e = \lambda \circ F^e$  with  $\lambda \in \text{Aut}(E)$ , deduce that  $[a] = ([1] + \lambda) \circ F^e$ , and conclude that  $q \mid a^2$ . So

$$a \equiv 0 \pmod{p} \iff q \mid a^2 \iff p^{\lceil e/2 \rceil} \mid a.$$

(This is a little stronger than the statement in [Sil09, Exercise 5.10(e)], and part (6) needs the stronger form.)

- (6) Show that no elliptic curve over  $\mathbb{F}_8$  has exactly 7 or exactly 11 rational points. Magma collects the values of  $\#E(\mathbb{F}_8)$  over all Weierstrass equations, and the corresponding traces:

```

K := GF(8); counts := {};
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([a1, a2, a3, a4, a6]);
  if ok then Include(~counts, #E); end if;
end for;
counts; // { 4, 5, 6, 8, 9, 10, 12, 13, 14 }
{ 9 - n : n in counts }; // { -5, -4, -3, -1, 0, 1, 3, 4, 5 }

```

Which integers in the Hasse interval are missing, and which traces belong to supersingular curves? Predict both from parts (1) and (5) before you look.

**Remark 1.3** (The traces of supersingular curves). The identity  $a^2 = \deg([1] + \lambda) \cdot q$  of Problem 1.2(5) says more than  $q \mid a^2$ . Since  $a^2 \leq 4q$ , the integer  $d := \deg([1] + \lambda)$  is at most 4, so a supersingular curve over  $\mathbb{F}_q$  has

$$a^2 \in \{0, q, 2q, 3q, 4q\},$$

and only the values with  $dq$  a perfect square can occur. Over  $\mathbb{F}_8$  these are  $a^2 \in \{0, 16\}$ , as found above; over  $\mathbb{F}_p$  with  $p \geq 5$  only  $a = 0$ ; over  $\mathbb{F}_3$  also  $a^2 = 9 = 3q$ , and over  $\mathbb{F}_2$  also  $a^2 = 4 = 2q$ , the cases of parts (3) and (4). Which of the five values occur, over which  $q$ , is a theorem of Waterhouse; we do not need it.

## 2. THE GROUP STRUCTURE OF $E(\mathbb{F}_q)$

**Fact 2.1** (Torsion and the Weil pairing). Let  $E$  be an elliptic curve over a field  $K$  and  $m \geq 1$  an integer with  $\text{char } K \nmid m$ .

- (1)  $E[m](\bar{K}) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$  [Sil09, Corollary III.6.4(b)].
- (2) If every  $m$ -torsion point of  $E(\bar{K})$  is  $K$ -rational, then  $K$  contains the  $m$ -th roots of unity [Sil09, Corollary III.8.1.1].

**Problem 2.2** (The group structure of  $E(\mathbb{F}_q)$ ).

This is [Sil09, Exercise 5.6]. Let  $E$  be an elliptic curve over  $\mathbb{F}_q$ .

- (1) Show that there are integers  $m, n \geq 1$  with  $p \nmid m$  and

$$E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}.$$

(Hint: decompose the finite abelian group  $E(\mathbb{F}_q)$  into its  $\ell$ -primary parts, and count the cyclic factors of each with Fact 2.1(1) for  $\ell \neq p$  and Fact 1.1(6) for  $\ell = p$ .)

- (2) With  $m$  as in part (1), show that  $q \equiv 1 \pmod{m}$ .
- (3) Suppose  $q = p \geq 5$  and  $E$  is supersingular. Show that  $m = 1$  or  $m = 2$ , and that  $m = 1$  if  $p \equiv 1 \pmod{4}$ .
- (4) Both values of  $m$  occur when  $p \equiv 3 \pmod{4}$ . Over  $\mathbb{F}_7$  the curves  $y^2 = x^3 + x$  and  $y^2 = x^3 - x$ , both with  $j = 1728$ , are supersingular; `Invariants(AbelianGroup(E))` returns  $[m, mn]$ , or  $[mn]$  when  $m = 1$ :

```
E1 := EllipticCurve([GF(7) | 1, 0]); E2 := EllipticCurve([GF(7) | -1, 0]);
IsSupersingular(E1), IsSupersingular(E2); // true true
Invariants(AbelianGroup(E1)); // [ 8 ]
Invariants(AbelianGroup(E2)); // [ 2, 4 ]
[ j : j in GF(13) | IsSupersingular(EllipticCurveFromjInvariant(j)) ]; // [ 5 ]
E3 := EllipticCurveFromjInvariant(GF(13)!5); aInvariants(E3); // [ 1, 0, 0, 7, 2 ]
Invariants(AbelianGroup(E3)); // [ 14 ]
Invariants(AbelianGroup(QuadraticTwist(E3))); // [ 14 ]
```

Explain the two values of  $m$  over  $\mathbb{F}_7$  from the rational points of order 2 of each curve, and the value over  $\mathbb{F}_{13}$  from part (3).

### 3. INFINITELY MANY ORDINARY PRIMES

Let  $E$  be an elliptic curve over  $\mathbb{Q}$ , given by a Weierstrass equation with coefficients in  $\mathbb{Z}$  and discriminant  $\Delta \in \mathbb{Z}$ . For a prime  $p \nmid \Delta$ , reducing the coefficients modulo  $p$  gives a Weierstrass equation over  $\mathbb{F}_p$  with nonzero discriminant, hence an elliptic curve  $E_p$  over  $\mathbb{F}_p$ ; we say that  $p$  is **supersingular** or **ordinary** for  $E$  according as  $E_p$  is. Silverman's Exercise 5.11 asks for infinitely many ordinary primes. The proof uses two facts we have not proved.

**Fact 3.1** (Reduction, and primes that split completely). Let  $E$ ,  $\Delta$  and  $E_p$  be as above.

- (1) Let  $p \nmid \Delta$  and let  $m \geq 1$  with  $p \nmid m$ . Every point of  $E(\mathbb{Q}_p)$  can be written  $[X : Y : Z]$  with  $X, Y, Z \in \mathbb{Z}_p$  not all divisible by  $p$ , and reducing the coordinates modulo  $p$  gives a group homomorphism  $E(\mathbb{Q}_p) \rightarrow E_p(\mathbb{F}_p)$  [Sil09, Proposition VII.2.1], whose restriction to the  $m$ -torsion points of  $E(\mathbb{Q}_p)$  is injective [Sil09, Proposition VII.3.1(b)].
- (2) Let  $L/\mathbb{Q}$  be a finite Galois extension. A prime  $p$  **splits completely** in  $L$  if  $p\mathcal{O}_L$  is a product of  $[L : \mathbb{Q}]$  distinct prime ideals of the ring of integers  $\mathcal{O}_L$ ; then for each prime  $\mathfrak{p}$  of  $L$  above  $p$  the completion  $L_{\mathfrak{p}}$  is  $\mathbb{Q}_p$ , so that  $L$  embeds in  $\mathbb{Q}_p$ . Infinitely many primes split completely in  $L$ : by the Chebotarev density theorem they have density  $1/[L : \mathbb{Q}]$  among all primes [Neu99].

**Problem 3.2** (Infinitely many ordinary primes).

This is [Sil09, Exercise 5.11], with “ordinary” for “Hasse invariant 1”. Fix a prime  $\ell \geq 3$ , and let  $L := \mathbb{Q}(E[\ell])$  be the field generated over  $\mathbb{Q}$  by the affine coordinates of the nonzero  $\ell$ -torsion points of  $E(\overline{\mathbb{Q}})$ .

- (1) Show that  $L$  is a finite Galois extension of  $\mathbb{Q}$ , and that  $L$  contains the  $\ell$ -th roots of unity.
- (2) Let  $p \nmid \ell\Delta$  be a prime that splits completely in  $L$ . Show that  $\ell^2 \mid \#E_p(\mathbb{F}_p)$  and that  $p \equiv 1 \pmod{\ell}$ .
- (3) Show that every prime  $p \geq 5$  with  $p \nmid \ell\Delta$  that splits completely in  $L$  is ordinary for  $E$ , and conclude. Where did you use  $\ell \geq 3$ ?
- (4) Take the curve of [Sil09, Example V.4.6],  $E: y^2 + y = x^3 - x^2 - 10x - 20$ , of discriminant  $-11^5$ , and  $\ell = 3$ . By part (2), a prime  $p \neq 3, 11$  that splits completely in  $\mathbb{Q}(E[3])$  has all nine 3-torsion points of  $E_p(\overline{\mathbb{F}}_p)$  rational. Magma lists the supersingular primes below 100 and the primes below 3000 with that property:

---

```

E := EllipticCurve([0, -1, 1, -10, -20]);           // y^2 + y = x^3 - x^2 - 10x - 20
Discriminant(E);                                   // -161051
good := [p : p in PrimesUpTo(3000) | p ne 11];
[p : p in good | p lt 100 and IsSupersingular(ChangeRing(E, GF(p)))];
// [ 2, 19, 29 ]
full := function(p, l)                             // are all the l-torsion points of E_p rational?
  I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
  return #I eq 2 and I[1] mod l eq 0;
end function;
split := [p : p in good | p ne 3 and full(p, 3)]; split;
// [ 337, 523, 1087, 2437, 2719, 2749 ]
[p mod 3 : p in split];                             // [ 1, 1, 1, 1, 1, 1 ]
[IsSupersingular(ChangeRing(E, GF(p))) : p in split];
// [ false, false, false, false, false, false ]
[p : p in good | p lt 300 and p ne 5 and full(p, 5)]; // and the same for l = 5:
// [ 31, 41, 61, 71, 101, 131, 151, 181, 191, 211, 241, 251, 271, 281 ]

```

---

Check the output against parts (2) and (3), and compare the list of supersingular primes with [Sil09, Example V.4.6]. The last line runs the same test for  $\ell = 5$ : which primes below 300 appear, and what does that suggest about the field  $\mathbb{Q}(E[5])$ ? (The rational point  $(5, 5)$  of  $E$  has order 5.)

**Remark 3.3** (Supersingular primes). The other direction is much harder. For the curve of Problem 3.2(4), D.H. Lehmer found exactly 27 odd supersingular primes below 31500 [Sil09, Example V.4.6]. For an elliptic curve over  $\mathbb{Q}$  without complex multiplication, the supersingular primes have density 0 (Serre, Elkies), and yet there are infinitely many of them (Elkies) [Sil09, Theorems V.4.7 and V.4.9]; the Lang–Trotter conjecture predicts that about  $c\sqrt{x}/\log x$  of them lie below  $x$  [Sil09, Conjecture V.4.8]. For a curve with complex multiplication, such as  $y^2 = x^3 + x$ , the supersingular primes have density 1/2 [Sil09, Examples V.4.4 and V.4.5].

## REFERENCES

- [Neu99] Jürgen Neukirch, *Algebraic number theory*, Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], vol. 322, Springer-Verlag, Berlin, 1999. Translated from the 1992 German original and with a note by Norbert Schappacher, With a foreword by G. Harder. MR1697859 ↑
- [Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑